

EXERCISE 5.2
Hypothesis Testing for 1 sample

[WEEK 9]



PROBABILITY & STATISTICAL DATA ANALYSIS

Exercise #1

A principal at a certain school claims that the students in his school are above average intelligence. A random sample of thirty students IQ scores have a mean score of 112. Is there sufficient evidence to support the principal's claim? The mean population IQ is 100 with a standard deviation of 15. Assume default significance level $\alpha=0.05$.

Answer

- Population mean, $\mu = 100$
- Population standard deviation, $\sigma = 15$
- Sample size, $n = 30$
- Sample mean, $\bar{x} = 112$
- Significance level, $\alpha = 0.05$

The principal claims that the students are **above average intelligence**, meaning:

$$\mu > 100$$

Since the population standard deviation is known, we use a **one-sample z-test**.

Step 1: State the Hypotheses

Null Hypothesis (H_0(

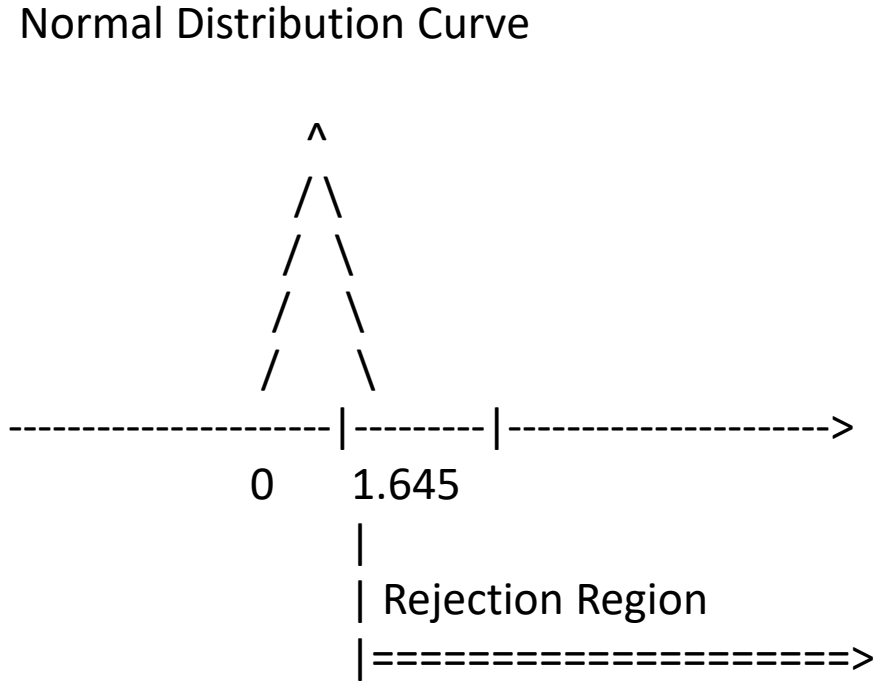
$$H_0: \mu = 100$$

Alternative Hypothesis (H_1(

$$H_1: \mu > 100$$

This is a **right-tailed test**.

- **Step 2: Identify the Critical Value**
- For:
- Right-tailed test
- $\alpha = 0.05$
- The critical z-value is:
- $z_{0.05} = 1.645$



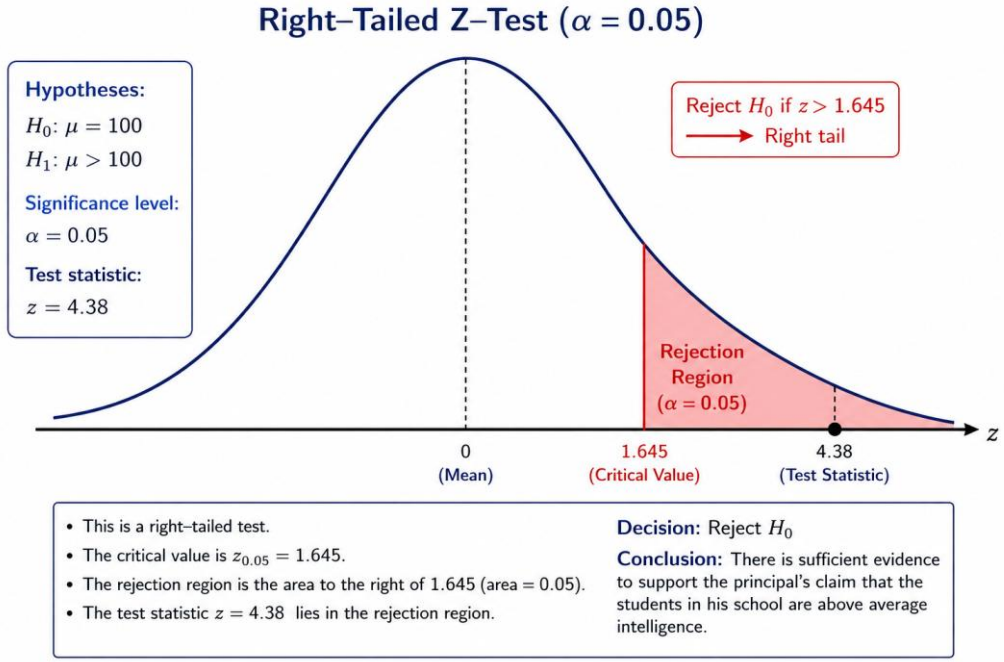
Step 3: Compute the Test Statistic

Use the z-test formula:

Substitute the values:

$$z = \frac{112 - 100}{15/\sqrt{30}}$$

z≈4.38



Compare with Critical Value

•Calculated z-value:

$$z = 4.38$$

•Critical value:

$$z_c = 1.645$$

Since:

$$4.38 > 1.645$$

the test statistic falls inside the rejection region.

Reject the null hypothesis H_0.

Exercise #2

A 2023 university report states that undergraduate students typically study an average of 20 hours per week. To verify this claim, a student newspaper conducted a survey among a random sample of 10 students. The survey found that students studied an average of 17.8 hours per week, with a sample standard deviation of 3.2 hours. Does this sample provide sufficient evidence at the 0.05 significance level to reject the claim that students study 20 hours a week?

H0:μ=20

$$H_1: \mu < 20$$

This is a **left-tailed one-sample t-test**.

Test statistic:

$$t = \frac{17.8 - 20}{3.2/\sqrt{10}} = -2.174$$

Degrees of freedom:

$$df = 9$$

Critical value at $\alpha = 0.05$:

$$t_{critical} = -1.833$$

Since:

$$-2.174 < -1.833$$

the test statistic falls in the **rejection region**.

p-value:

$$p = 0.0289$$

Since:

$$0.0289 < 0.05$$

Final conclusion

At the 0.05 significance level, there is **sufficient evidence to reject the claim** that students study an average of 20 hours per week.

The data supports that students study **less than 20 hours per week**.

